### v. Stay insulator



Fig. 1.16 **Stay insulator**

The insulator used in the stay wire is called a stay insulator. This type of insulator is as shown in figure 1.16 is made of porcelain. The size of insulator depends upon the tensile strength of stay wire. It is used to prevent the leakages from the poles up to 33KV. These are provided at a height of 3m from the ground level.

## 1.10 Power distribution system

The medium voltage transformer using transmission lines are reduced by step down transformer (415V, or 240V) are used to the consumer. This system includes the feeder lines, distributors and service mains.

### 1.10.1 Feeder line

The feeder line is nothing but similar as power transmission conductors. It connects the power from sub-station and power supply to the consumers. The voltage is uniform throughout the conductor.

### 1.10.2 Distributor

Electric power is distributed to the consumer by tapings. Since electric power is taken from many tappings and the current is not uniform in all. Therefore, when designing the distribution system, care must be taken to the voltage drop.

### 1.10.3 Classification of distribution system

According to scheme of connections the distribution system is classified as,

a. Radial distribution system
b. Ring main distribution system
c. Grid or interconnected distribution system

### a. Radial distribution system

In this system, each load junctions are connected through separate feeders and they are controlled by the sub-stations. This method is used when the low voltage exists in the center of the city. If there is a fault in any feeder, the whole circuit will be affected.

### b. Ring main distribution system

This system designed like a closed ring. Each load junctions were constructed one after another. Electricity is provided from two supply sources in different places for each load junctions. If one gets faulted, the other can be used to get power supply. This system is used in places where low and medium voltage is required. In this way there may be a chance of low voltage fluctuations happening for the consumers. Power supply can be provided by more than one feeder. Reliability can be generated by supply of electricity through each of the two feeders.

### c. Grid or interconnected distribution system

More than one power plants and sub-stations are connected in series feeders are called as 'Interconnected distribution



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